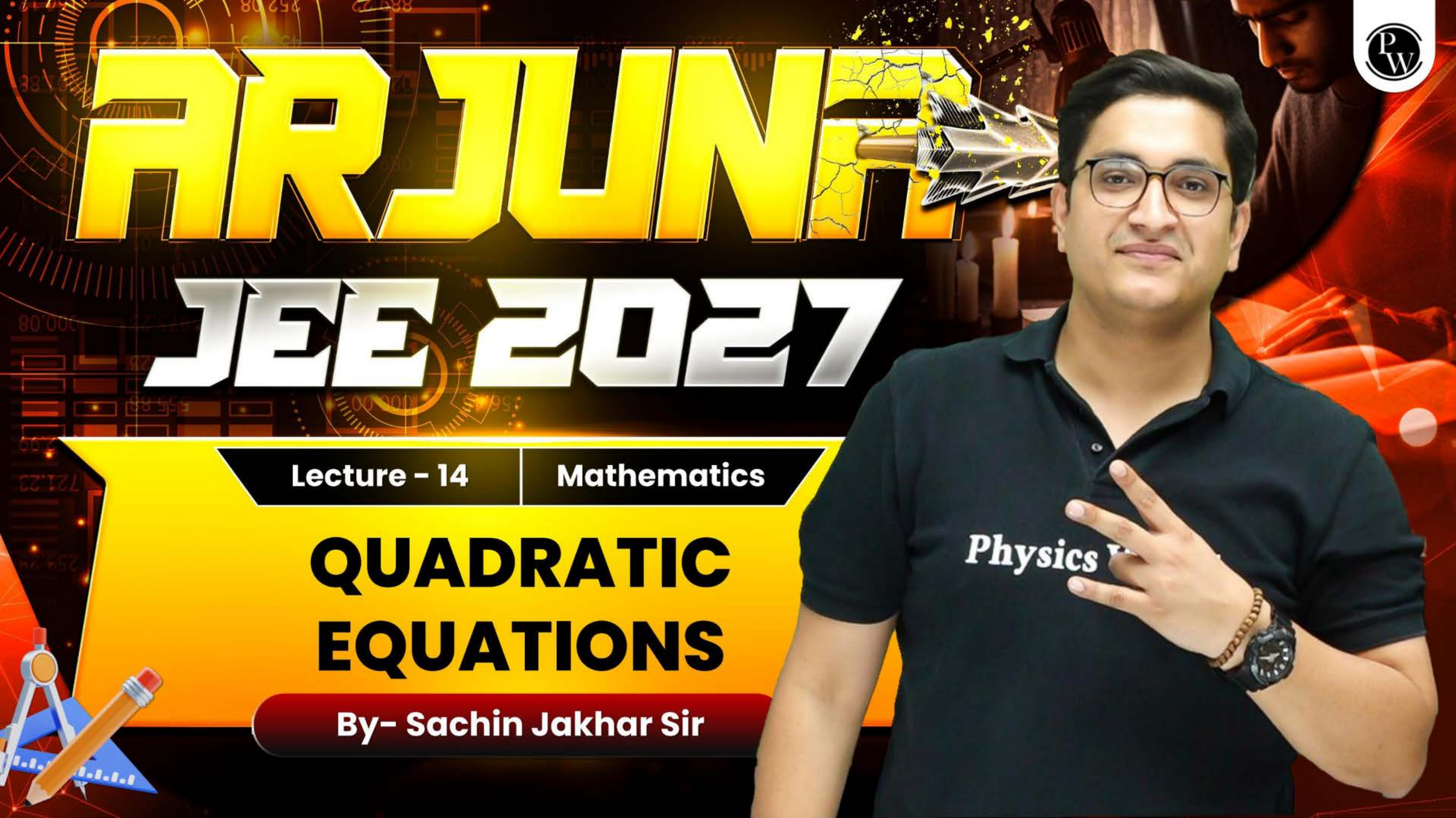
ARJUNA JEE 2027

Lecture - 14 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

- **A** General Second Degree Equation
- **B** Equations Reducible to Quadratic
- **C** Questions on Polynomials
- **D** Question Practice



SECOND DEGREE EQUATION

$$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$$

Condition that it can be resolved as product of two linear factors is :

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

RAR

OR

$$\# \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

linear factor

Variables

x or y

Range!

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$$\begin{vmatrix} +a & -h & +g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$$+a\begin{vmatrix} b & f \\ f & c \end{vmatrix} - h\begin{vmatrix} h & f \\ g & c \end{vmatrix} + g\begin{vmatrix} h & b \\ g & f \end{vmatrix}$$

$$a(bc-f^2) - h(hc-gf) + g(hf-bg)$$

$$abc - af^2 = ch^2 + hgf$$

$$+ hgf - bg^2 = 0$$

$$\boxed{2hgf + abc - af^2 - ch^2 - bg^2 = 0}$$

Determinant

- as a Tool
- Coordinate
- Cross-product

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

3×3:

$$\begin{vmatrix} 2 & 1 & -1 \\ 3 & 1 & 4 \\ -2 & 1 & -3 \end{vmatrix} = -18$$

$$= +2\begin{vmatrix} 1 & 4 \\ 1 & -3 \end{vmatrix} - 1\begin{vmatrix} 3 & 4 \\ -2 & -3 \end{vmatrix} + (-1)\begin{vmatrix} 3 & 1 \\ -2 & 1 \end{vmatrix}$$

$$= 2(-3-4) - 1(-9+8) - 1(3-(-2))$$

$$= 2(-7) + 1 - 5 = -14 + 1 - 5$$

$$= -18$$

$$\# \begin{vmatrix} -2 & -3 & 1 \\ 1 & 0 & 5 \\ -4 & 1 & 6 \end{vmatrix} = 89.$$

$$+(-2)\begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} - (-3)\begin{vmatrix} 1 & 5 \\ -4 & 6 \end{vmatrix} + 1\begin{vmatrix} 1 & 0 \\ -4 & 1 \end{vmatrix}$$

$$-2(0-5) + 3(6-(-20)) + 1(1-0)$$

$$10 + 3 \times 26 + 1$$

$$10 + 78 + 1 = 89$$

$$\# \begin{vmatrix} 1 & -5 & -4 \\ 3 & 2 & 1 \\ 0 & -1 & 6 \end{vmatrix} = ?$$

Chat - DIBY - X_1

QUESTION

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

Find value of k for which $12x^2 - 10xy + 2y^2 + 11x - 5y + k$ is a product of two linear factors. Hence find those factors.

$a = 12$

$b = 2$

$2h = -10 \Rightarrow h = -5$

$2g = 11 \Rightarrow g = \frac{11}{2}$

$2f = -5 \Rightarrow f = -\frac{5}{2}$

$c = k$

$$\# \begin{vmatrix} 12 & -5 & \frac{11}{2} \\ -5 & 2 & -\frac{5}{2} \\ \frac{11}{2} & -\frac{5}{2} & k \end{vmatrix} = 0 \xrightarrow{\text{DIBY}} k = 2$$

QUESTION

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

Find value of k for which $12x^2 - 10xy + 2y^2 + 11x - 5y + k$ is a product of two linear factors. Hence find those factors.

$$12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$$

$$12x^2 + (11 - 10y)x + 2y^2 - 5y + 2 = 0$$

A = 12, B = $11 - 10y$, C = $2y^2 - 5y + 2$

$$x = \frac{-(11-10y) \pm \sqrt{(11-10y)^2 - 4 \times 12 (2y^2 - 5y + 2)}}{2(12)}$$

$$= \sqrt{(2y+5)^2} = |2y+5|$$

$$121 + 100y^2 - 220y - 48(2y^2 - 5y + 2)$$

$$121 + 100y^2 - 220y - 96y^2 + 240y - 96$$

$$4y^2 + 20y + 25$$

$x^1 = -\dfrac{(11-10y^1) \pm (2y^1+5)}{24}$

$\oplus$

$$x = \frac{10y - 11 + 2y + 5}{24}$$

$$24x = 12y - 6$$

$$4x = 2y - 1$$

$$\boxed{4x - 2y + 1 = 0}$$

$\ominus$

$$x = \frac{10y - 11 - 2y - 5}{24}$$

$$24x = 8y - 16$$

$\div 8$

$$3x = y - 2$$

$$\boxed{3x - y + 2 = 0}$$

Given.

$$(4x - 2y + 1)(3x - y + 2) = 0$$

✓

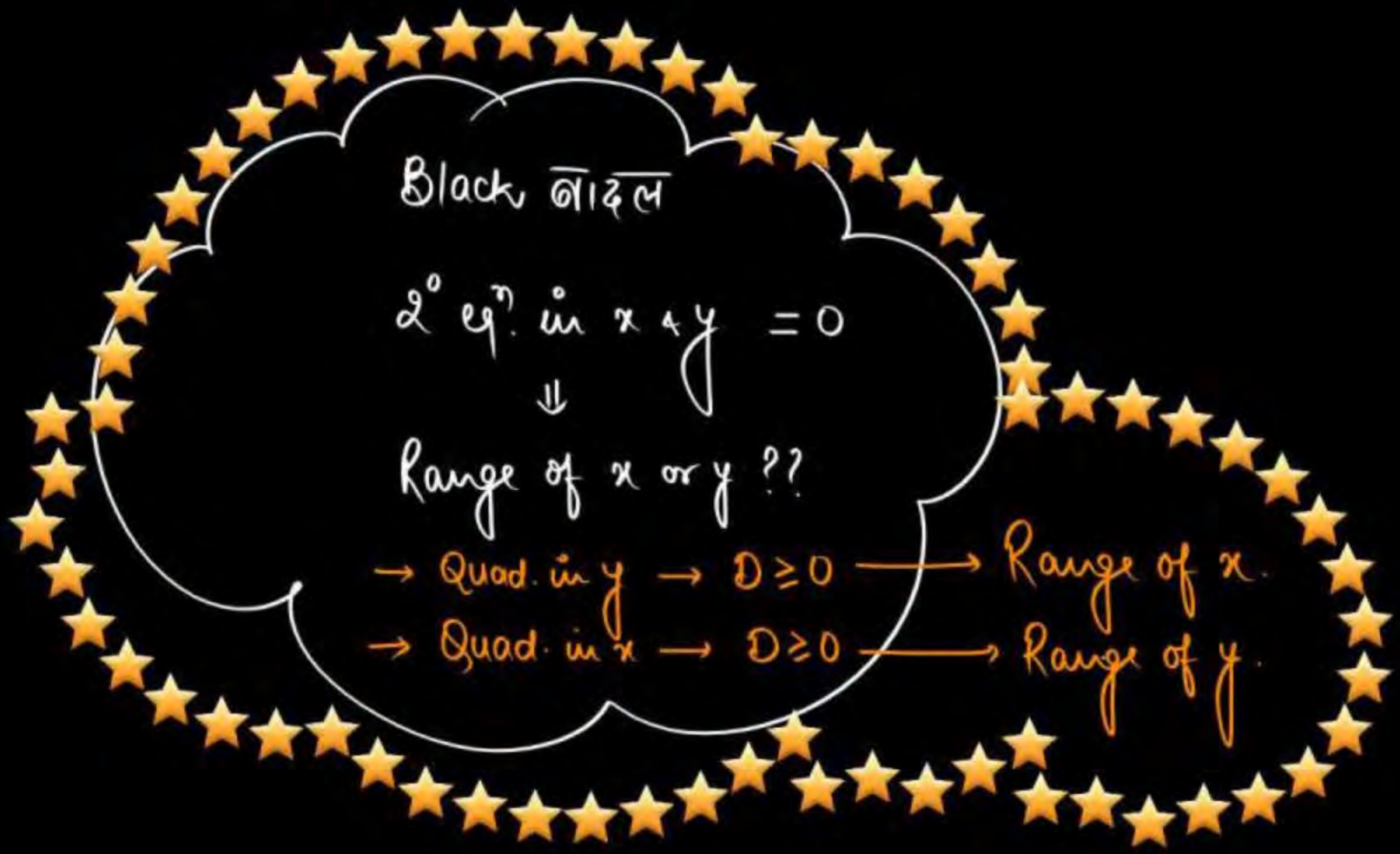
Ques:-

$$3x^2 - y^2 + 4xy - 6x - 8y + \alpha = 0$$

find $\alpha \rightarrow$ product of 2 linear factors.

find those factors also!

DIBY - (X2)


Black बादल
2° eq^n. in x & y = 0
⇓
Range of x or y ??
→ Quad. in y → D ≥ 0 → Range of x.
→ Quad. in x → D ≥ 0 → Range of y.

QUESTION

If x, y be real and $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, show that $x \in [3, 6], y \in [1, 10]$.

∘ Quad. in x :

$$\overset{A}{9}x^2 + \overset{B}{2(y-46)}x + \overset{C}{(y^2-20y+244)} = 0$$

Since x is a Real no. $\Rightarrow$ $D \geq 0$

$$4(y-46)^2 - 4 \times 9 \times (y^2-20y+244) \geq 0$$

$$y^2 + 46^2 - 92y - 9y^2 + 180y - 9 \times 244 \geq 0$$

$$-8y^2 + 88y + 23 \times 23 \times 4 - 9 \times 244 \geq 0$$

$\div 4$

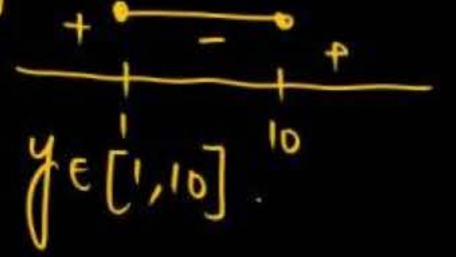
$$-2y^2 + 22y + 23^2 - 9 \times 61 \geq 0$$

$$-2y^2 + 22y - 20 \geq 0$$

$\div (-2)$

$$y^2 - 11y + 10 \leq 0$$

$$(y-1)(y-10) \leq 0$$



$y \in [1, 10]$.

QUESTION

If x, y be real and $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, show that $x \in [3, 6], y \in [1, 10]$.

• Quad. in y:

$$\underset{A}{1}\,y^2 + \underset{B}{2(x-10)}\,y + \overbrace{(9x^2 - 92x + 244)}^{C} = 0.$$

$$D \geq 0.$$

$$4(x-10)^2 - 4 \times 1\,(9x^2 - 92x + 244) \geq 0$$

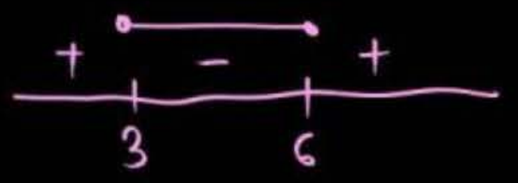
$$x^2 + 100 - 20x - 9x^2 + 92x - 244 \geq 0$$

$$-8x^2 + 72x - 144 \geq 0.$$

$$\div 8$$

$$x^2 - 9x + 18 \leq 0.$$

$$(x-6)(x-3) \leq 0$$



$$x \in [3, 6]$$

QUESTION [JEE MAINS-2021]

If $x^2 + 9y^2 - 4x + 3 = 0, x, y \in R$, then x and y respectively lie in the intervals.

(A) [1, 3] and [1, 3]

(B) [1, 3] and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

(C) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and [1, 3]

(D) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

DIBY – (L)

$x^2 + 9y^2 + 0xy - 4x + 0y + 3 = 0.$

QUESTION

If x, y and z are three real numbers such that $x + y + z = 4$ and $x^2 + y^2 + z^2 = 6$, then show that each of x, y and z lie in the closed interval $\left[\frac{2}{3}, 2\right]$.

$z = 4 - (x + y)$

$x^2 + y^2 + (4 - x - y)^2 = 6$

$x^2 + y^2 + 16 + x^2 + y^2 - 8x + 2xy - 8y - 6 = 0$

$2x^2 + 2y^2 + 2xy - 8x - 8y + 10 = 0$

$x^2 + y^2 + xy - 4x - 4y + 5 = 0$

$y^2 + (x - 4)y + (x^2 - 4x + 5) = 0$

$D \geq 0$

$(x-4)^2 - 4 \times (x^2 - 4x + 5) \geq 0$

$x^2 + 16 - 8x - 4x^2 + 16x - 20 \geq 0$

$-3x^2 + 8x - 4 \geq 0$

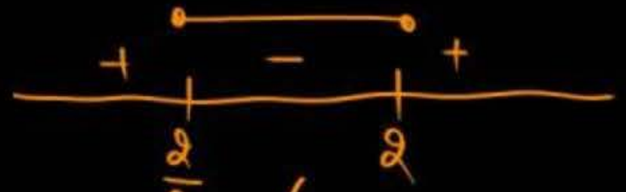
$$-3x^2+8x-4 \geq 0$$

$\ominus$

$$3x^2-8x+4 \leq 0$$

$$3x^2-6x-2x+4 \leq 0$$

$$(3x-2)(x-2) \leq 0$$



$$x \in \left[\frac{2}{3}, 2\right]$$

HHPP.

EQUATIONS REDUCIBLE TO QUADRATIC

Format –I : $(x-a)(x-b)(x-c)(x-d)=k$

format ⇓ Action !!

Ex. : $(x+1)(x+2)(x+3)(x+4)=\frac{9}{16}$

$(x+1)(x+4)\ (x+2)(x+3) = \frac{9}{16}$

multiply any two brackets such that variable part is same.

$(x^2+5x+4)(x^2+5x+6) = \frac{9}{16}$

$x^2+5x+4 = t \Rightarrow t(t+2) = \frac{9}{16}$

OR

$x^2+5x = t \Rightarrow (t+4)(t+6) = \frac{9}{16}$

as (t)

Quad. in (t)

DIBY.

EQUATIONS REDUCIBLE TO QUADRATIC

Format –II : $(x-a)(x-b)(x-c)(x-d) = kx^2$

Ex. : $(x+2)(x+3)(x+8)(x+12) = 4x^2$

aise 2 brackets ki ko multiply karo! ^ const. part → same.

$(x+2)(x+12)\ (x+3)(x+8) = 4x^2$.

$\Rightarrow \left(x^2+14x+24\right)\left(x^2+11x+24\right) = 4$

$\left(x+14+\frac{24}{x}\right)\left(x+11+\frac{24}{x}\right) = 4$.

$\left(\underbrace{x+\frac{24}{x}}_{t}+14\right)\left(\underbrace{x+\frac{24}{x}}_{t}+11\right) = 4$.

$(t+3)$ (t)

$t(t+3) = 4$

↳ Quad.

EQUATIONS REDUCIBLE TO QUADRATIC

Format –III : $(x-a)^4 + (x-b)^4 = k$

Ex. : $(x-6)^4 + (x-8)^4 = 16$

Put: $y = \frac{x-a+x-b}{2}$

$y = \frac{x-6+x-8}{2}$

$2y = 2x - 14$

$y = x - 7$

$y + 7 = x$

put: $x = y + 7$.

$(y+7-6)^4 + (y+7-8)^4 = 16$.

$(y+1)^4 + (y-1)^4 = 16$

$2y^4 + 12y^2 + 2 = 16$.

$2y^4 + 12y^2 - 14 = 0$

$y^4 + 6y^2 - 7 = 0$

$y^2 = t$.

Quad. $t^2 + 6t - 7 = 0$

$$(y+1)^4 = ((y+1)^2)^2$$

$$= (y^2+2y+1)^2$$

$$(y+1)^4 = y^4 + 4y^2 + 1 + 4y^3 + 4y + 2y^2.$$

$$(y-1)^4 = ((y-1)^2)^2 = (y^2-2y+1)^2$$

$$(y-1)^4 = y^4 + 4y^2 + 1 - 4y^3 - 4y + 2y^2.$$

add.

$$(y+1)^4 + (y-1)^4 = 2y^4 + 8y^2 + 2 + 4y^2$$

$$= 2y^4 + 12y^2 + 2.$$

EQUATIONS REDUCIBLE TO QUADRATIC

Format –IV : Reciprocal Equation

$$ax^3 + bx^2 + bx + a = 0$$
$$ax^4 + bx^3 + cx^2 + bx + a = 0$$
$$ax^5 + bx^4 + cx^3 + cx^2 + bx + a = 0$$

100% Ques? आएगा!!

even degree wali

$ax^4+bx^3+cx^2+bx+a=0$ → divide by x^2

$ax^2+bx+a=0$ → divide by x

$ax^6+bx^5+cx^4+dx^3+cx^2+bx+a=0$ → divide by x^3

odd degree wali

$ax^3+bx^2+bx+a=0$

$ax^5+bx^4+cx^3+cx^2+bx+a=0$

RRT

$x=1$ or -1 → ek Root

→ divide by $(x\mp1)$ ⇓ even degree recip. eqn

EXAMPLE

Solve:

(i) $2x^4 + x^3 - 11x^2 + x + 2 = 0$

(ii) $6x^5 + 11x^4 - 33x^3 - 33x^2 + 11x + 6 = 0$

(iii) $12x^4 - 56x^3 + 89x^2 - 56x + 12 = 0$

divide by x^2

$$2x^2 + x - 11 + \frac{1}{x} + \frac{2}{x^2} = 0$$

$$2\left(x^2 + \frac{1}{x^2}\right) + \left(x + \frac{1}{x}\right) - 11 = 0$$

put: $x + \frac{1}{x} = t$

SBS. $x^2 + \frac{1}{x^2} + 2 = t$

$$2(t^2 - 2) + t - 11 = 0$$

$$2t^2 + t - 15 = 0$$

Quad. in (t)

Basic Maths.

"Term + uska reciprocal"

$$\left(x-\frac{1}{x}\right)^2+2 = x^2+\frac{1}{x^2} = \left(x+\frac{1}{x}\right)^2-2$$

EXAMPLE

Solve:

(i) $2x^4 + x^3 - 11x^2 + x + 2 = 0$

(ii) $6x^5 + 11x^4 - 33x^3 - 33x^2 + 11x + 6 = 0$

(iii) $12x^4 - 56x^3 + 89x^2 - 56x + 12 = 0$

$\div x^2$

$12x^2 - 56x + 89 - \frac{56}{x} + \frac{12}{x^2} = 0$

$12\left(x^2 + \frac{1}{x^2}\right) - 56\left(x + \frac{1}{x}\right) + 89 = 0$

$12\left(t^2 - 2\right) - 56t + 89 = 0$

$x = -1$

$-6 + 11 + 33 - 33 - 11 + 6 = 0$.

odd degree

$x = -1$

	6	11	-33	-33	11	6
		-6	-5	38	-5	-6
	6	5	-38	5	6	0

$(x+1)\left(6x^4 + 5x^3 - 38x^2 + 5x + 6\right) = 0$

even degree wali recip. eq^n

$\div x^2$

QUESTION [JEE Mains-2023]

The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0$, $x \in R$ has

(A) Two solutions and both are negative

(B) No solution

(C) Four solutions two of which are negative

(D) Two solutions and only one of them is negative

$e^x = t$

$(e^x)^4 = t^4$

$t^4 + 8t^3 + 13t^2 - 8t + 1 = 0$

$\div t^2$

$t^2 + 8t + 13 - \frac{8}{t} + \frac{1}{t^2} = 0$

$\left(t^2 + \frac{1}{t^2}\right) + 8\left(t - \frac{1}{t}\right) + 13 = 0$

$t - \frac{1}{t} = p$

SBS $t^2 + \frac{1}{t^2} - 2 = p^2$

$(p^2 + 2) + 8p + 13 = 0$

$\# \; \beta^2 + 8\beta + 15 = 0$

$(\beta+3)(\beta+5) = 0$

$\beta = -3 \text{ or } -5$

$t - \frac{1}{t} = -3$

$t^2 - 1 = -3t$

$t^2 + 3t - 1 = 0$

$t = \frac{-3 \pm \sqrt{13}}{2}$

$\# \; e^x = \frac{-3+\sqrt{13}}{2}$

$\# \; e^x = \frac{-3-\sqrt{13}}{2} \longrightarrow$ Rejected !!

e^x : +ve always.

$\frac{-3-\sqrt{13}}{2}$: -ve.

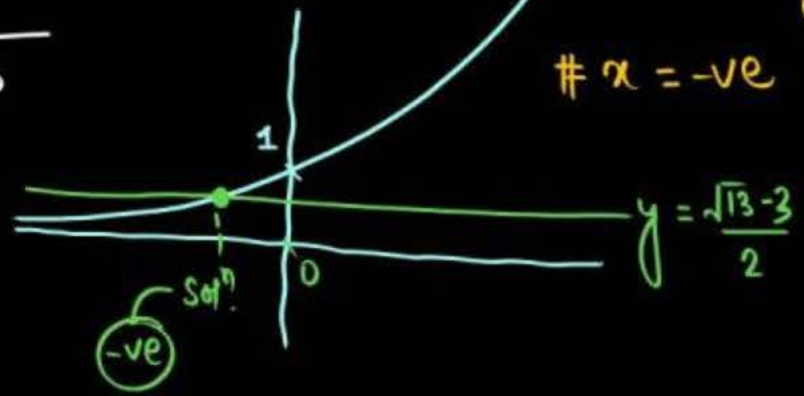
$e^x = \frac{\sqrt{13}-3}{2}$

$\log_e(e^x) = \log_e\left(\frac{\sqrt{13}-3}{2}\right)$

$x = \log_e\left(\frac{\sqrt{13}-3}{2}\right)$

$\frac{\sqrt{13}-3}{2}$: 0.xxx

$\# \; x = -ve$



$p^2 + 8p + 15 = 0$

$(p+3)(p+5) = 0$

$p = -3$ or -5

$t - \frac{1}{t} = -5$

$t^2 + 5t - 1 = 0$

$t = \frac{-5 \pm \sqrt{29}}{2}$

$e^x = \frac{-5 - \sqrt{29}}{2}$ ✗

$e^x = \frac{-5 + \sqrt{29}}{2}$

$e^x = \frac{\sqrt{29} - 5}{2}$

$x = -ve$

QUESTION [JEE Mains-2022]

The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ is _____.

Ans: 02

DIBY - χ_3

EQUATIONS REDUCIBLE TO QUADRATIC

Format –V : Infinite Repetition

first rep. → Re-substitute

ex :- $x = \sqrt{5\sqrt{5\sqrt{5\sqrt{5\ldots\infty}}}} = 5$

$\sqrt{5\sqrt{5\sqrt{5\sqrt{5\ldots\infty}}}} \rightarrow x$.

\# $x = \sqrt{5x}$

SBS:

$x^2 = 5x$

$x(x-5) = 0$

$x = 0, 5$ ✓ ✗

ex :- $\frac{1+\sqrt{29}}{2} = \alpha = \sqrt{7+\sqrt{7+\sqrt{7+\sqrt{7+\ldots\infty}}}}$, $\sqrt{7+\sqrt{7+\sqrt{7+\ldots\infty}}} \rightarrow \alpha$.

$\alpha = \sqrt{7+\alpha}$

SBS: $\alpha^2 = 7+\alpha$

$\alpha^2 - \alpha - 7 = 0$

$\alpha = \frac{1 \pm \sqrt{1-4(-7)}}{2}$

$\alpha = \frac{1 \pm \sqrt{29}}{2}$

$\frac{1+\sqrt{29}}{2}$ ✓

$\frac{1-\sqrt{29}}{2}$ ✗

QUESTION [IIT-JEE-2012]

The value of $6+\log_{\frac{3}{2}}\left(\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\ldots}}}\right)$ is

$\alpha = \frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\ldots}}}$ $\rightarrow \alpha$.

$6+\log_{\frac{3}{2}}\left(\frac{4}{9}\right) = 6-2 = 4$ Ans. (-2)

$\alpha = \frac{1}{3\sqrt{2}}\sqrt{4-\alpha}$

SBS: $\alpha^2 = \frac{1}{18}(4-\alpha)$

$18\alpha^2+\alpha-4=0$

$18\alpha^2+9\alpha-8\alpha-4=0$

$9\alpha(2\alpha+1)-4(2\alpha+1)=0$

$(9\alpha-4)(2\alpha+1)=0 \rightarrow \alpha = -\frac{1}{2}, \frac{4}{9}$

$-\frac{1}{2}$ ✗ , $\frac{4}{9}$ ✓

QUESTION [JEE Mains-2021]

$2-\frac{2}{5}\sqrt{30}$ ✗

The value of $4+\cfrac{1}{5+\cfrac{1}{4+\cfrac{1}{5+\cfrac{1}{4+\cdots\infty}}}}$ is

(A) $2+\frac{2}{5}\sqrt{30}$

(B) $2+\frac{4}{\sqrt{5}}\sqrt{30}$

(C) $4+\frac{4}{\sqrt{5}}\sqrt{30}$

(D) $5+\frac{2}{5}\sqrt{30}$

$$x = 4+\cfrac{1}{5+\cfrac{1}{\underbrace{4+\cfrac{1}{5+\cfrac{1}{4+\cdots\infty}}}_{x}}}$$

$$\Rightarrow x = 4+\frac{1}{5+\frac{1}{x}}$$

$$x = 4+\frac{x}{5x+1}$$

$$(x-4) = \frac{x}{5x+1}$$

$$(x-4)(5x+1) = x$$

$$5x^2-20x+x-4 = x$$

$$5x^2-20x-4 = 0$$

$$x = \frac{20 \pm \sqrt{16\times 30}}{10}$$

$$= 2 \pm \frac{4}{10}\sqrt{30}$$

$$x = 2 \pm \frac{2}{5}\sqrt{30}$$ ✓

$x = 2+\frac{2}{5}\sqrt{30}$ ✗

QUESTION [JEE Mains-2021]

The value of $3+\cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+\ldots\infty}}}}$ is

(A) $1.5+\sqrt{3}$

(B) $2+\sqrt{3}$

(C) $3+2\sqrt{3}$

(D) $4+\sqrt{3}$

DIBY - x_4

FACTOR AND REMAINDER THEOREM

Factor Theorem :

If $(x - a)$ is a factor of $P(x)$, then $P(a) = 0$.

Polynomial

$x = a$

ex:- $P(x) = x^2 - 5x + 6$

$x - 2 = 0$ is a factor

$x = 2$

$P(2) = 0$.

Remainder Theorem :

If polynomial P(x) is divided by $(x - a)$ then Remainder = $P(a)$

$$
\begin{array}{r|l}
 & x - 1 \\
\hline
x - 4 & x^2 - 5x + 6 \\
 & x^2 - 4x \\
 & \hline
 & -x + 6 \\
 & -x + 4 \\
 & \hline
 & R = 2
\end{array}
$$

ex:- $P(x) = x^2 - 5x + 6$

divided by $(x-4)$, R = ??

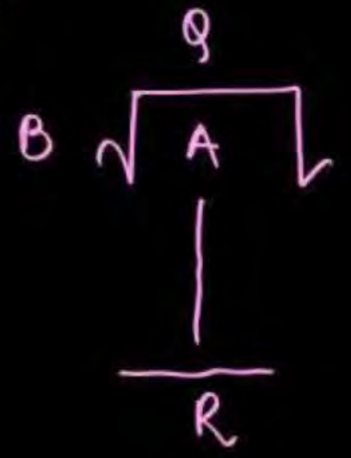
$x - 4 = 0$

$x = 4$

$R = P(4) = 16 - 20 + 6$

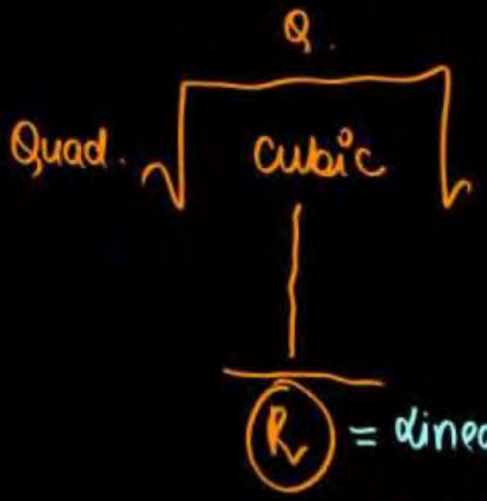
$= 2$.

#



① $\Rightarrow$ $A = BQ + R$ (always True)

② # max. degree of Rem. < degree of B



R = linear / const.

$= ax + b$.

EXAMPLE

Find the remainder when

(i) $x^{100} + x + 2$ **is divided by** $x - 1$

(ii) $x^{100} + x + 2$ **is divided by** $x^2 - 1$

$P(x) = x^{100} + x + 2$

div. $(x-1) = 0 \rightarrow x = 1$

$R = P(1) = 1^{100} + 1 + 2 = 4$

$Q(x)$

$(x^2-1) \overline{)\ x^{100} + x + 2}$

Quad.

$R = ax + b$.

$x^{100} + x + 2 = (x^2-1)\,Q(x) + (ax+b)$ (id.)

$x = 1$: $4 = 0 + a + b$

$x = -1$: $2 = -a + b$

$\left.\begin{array}{l} a+b=4 \\ -a+b=2 \end{array}\right\}$

$2b = 6$

$b = 3,\ a = 1.$

$R(x) = x + 3.$

EXAMPLE

$$\# \quad R(x) = \frac{7}{6}x^2 + \frac{3}{2}x - \frac{2}{3}$$

A polynomial in x of degree greater than 3 leaves the remainder 2, 1 and -1 when divided by $(x-1); (x+2)$ & $(x+1)$ respectively. Find the remainder, if the polynomial is divided by, $(x^2-1)(x+2)$.

degree $(P(x)) > 3$.

$$(x-1)\overline{)P(x)} \quad Q_1(x) \quad \text{Rem} = 2$$

$$(x+2)\overline{)P(x)} \quad Q_2(x) \quad \text{Rem} = 1$$

$$(x+1)\overline{)P(x)} \quad Q_3(x) \quad \text{Rem} = -1$$

$x=1$: $P(x) = (x-1)Q_1(x) + 2$ $\Rightarrow$ $P(1) = 2$.

$P(-2) = 1$.

$P(-1) = -1$

Cubic $\rightarrow$ $(x^2-1)(x+2)\overline{)P(x)}$ $Q(x)$ $\quad R = ax^2+bx+c$.

(degree = 3)

$$P(x) = (x^2-1)(x+2)\,Q(x) + ax^2+bx+c.$$

$x=1 \Rightarrow P(1) = a+b+c = 2$

$x=-1 \Rightarrow P(-1) = a-b+c = -1$

$x=-2 \Rightarrow P(-2) = 4a-2b+c = 1$

$4a+c = 4$

$a+c = \frac{1}{2}$

$3a = 4 - \frac{1}{2}$

$3a = \frac{7}{2}$

$2b = 3$

$b = \frac{3}{2}$

$a = \frac{7}{6}$ $\quad c = -\frac{2}{3}$

CHALLENGER QUESTION



New to Me!!

If $p(x)$ is polynomial of degree five and $p(0) = -1, p(1) = 1, p(2) = 4, p(3) = 9, p(4) = 16, p(5) = 25$, then find the value of $p(6) =$ _____.

→ wrong ans: (36)

By - pattern

$p(x) = x^2$, for $x = 1,2,3,4,5$

$p(x) - x^2 = 0$

- dong drive

$$p(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f$$

→ 6 - cond^n & 6 - unknown eq^n.

$g(x) = p(x) - x^2 = 0$

degree = 5 → Roots = 5

$1 = x$

$2 = x$

$3 = x$

$4 = x$

$5 = x$

$g(x) = p(x) - x^2 = a(x-1)(x-2)(x-3)(x-4)(x-5)$ — id.

$x = 0$

$p(0) - 0 = a(-1)(-2)(-3)(-4)(-5)$

$1 = 120a$

$a = \frac{1}{120}$.

$p(6) = 37$ — fnf.

$p(x) - x^2 = \frac{1}{120}(x-1)(x-2)(x-3)(x-4)(x-5)$

$x = 6$

$p(6) - 36 = \frac{1}{120}(5)(4)(3)(2)(1) = 1$ (120)

Ques:-

$p(x)$ is a **monic polynomial** → polynomial whose leading coeff = 1.

of degree = 4

& $p(2) = 5$, $p(3) = 7$, $p(4) = 9$, $p(7) = 15$

$= 2\times2+1$ $\quad 2\times3+1$ $\quad 2\times4+1$ $\quad 2\times7+1$

Then find $p(8) = ??$

Soln:

$p(x) = x^4 + ax^3 + bx^2 + cx + d$

(X) long drive!

Soln:

$p(x) = 2x+1$ only

$x = 2, 3, 4, 7$.

$g(x) = p(x) - (2x+1) = 0$ → 2, 3, 4, 7

degree = 4

$\# \quad p(x) - (2x+1) = 1(x-2)(x-3)(x-4)(x-7)$

$x = 8$

$p(8) - 17 = 6\times5\times4\times1$.

$p(8) = 120 + 17 = 137$ Ans.

QUESTION [JEE Mains-2021]

Let $f(x)$ be a polynomial of degree 3 such that $f(k) = \frac{-2}{k}$ for $k = 2, 3, 4, 5$. Then the value of $52 - 10f(10)$ is equal to

$52 - 26 = 26$

$f(x)$

$* \quad f(x) = -\frac{2}{x}$ — $2, 3, 4, 5$

$f(x) + \frac{2}{x} = 0$

$g(x) = x f(x) + 2 = a(x-2)(x-3)(x-4)(x-5)$

degree = 4 (of $g(x)$); degree = 3 (of $x f(x)$)

$x = 0$

$0 f(0) + 2 = a(-2)(-3)(-4)(-5)$

$2 = 120a \Rightarrow a = \frac{1}{60}$

$x f(x) + 2 = \frac{1}{60}(x-2)(x-3)(x-4)(x-5)$

$x = 10$

$10 f(10) + 2 = \frac{1}{60}(8)(7)(6)(5)$

$10 f(10) = 28 - 2 = 26$

QUESTION

If $f(x)$ is a polynomial of degree four with leading coefficient one satisfying $f(1) = 1, f(2) = 2, f(3) = 3$, then find $\left[\frac{f(-1)+f(5)}{f(0)+f(4)}\right]$ (where [.] represents greatest integer)

Ans: 5

DIBY - X_5

It is never about end result

It's always about Journey

Thank You

#futureIITians